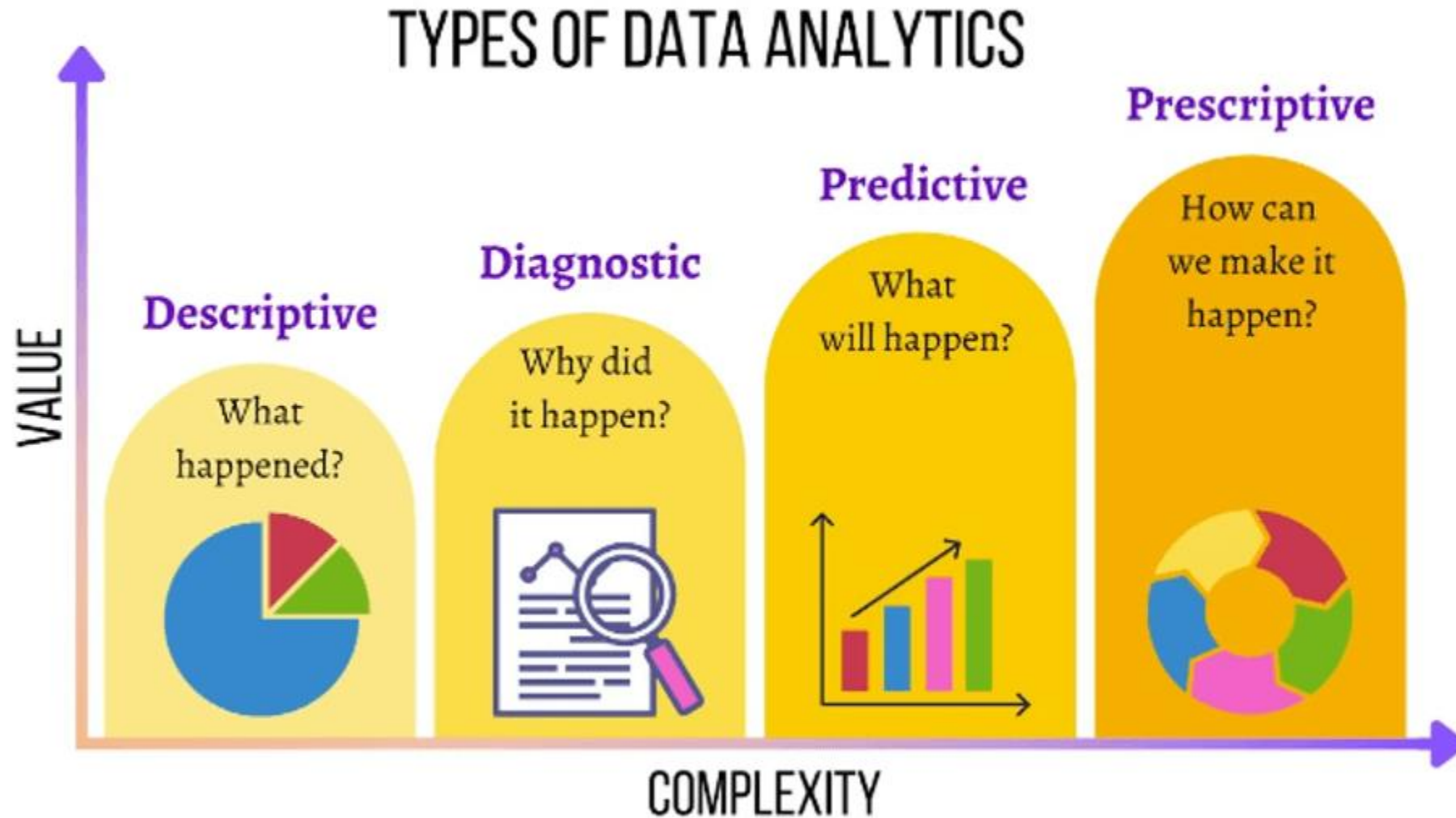


❖ DIFFERENT TYPES OF DATA ANALYSIS

❖ The four main types of data analysis: Descriptive, Diagnostic, Predictive, and Prescriptive



❖ **Descriptive Analytics**

- ❖ **Descriptive analytics is a simple, surface-level type of analysis that looks at what has happened in the past.**
- ❖ The two main techniques used in descriptive analytics are **data aggregation and data mining**—so, the data analyst **first gathers the data and presents** it in a summarized format (that's the aggregation part) and then **“mines” the data to discover patterns**.
- ❖ The data is then presented in a way that can be easily understood by a wide audience (not just data experts).
- ❖ It's important to note that descriptive analytics doesn't try to explain the historical data or establish cause-and-effect relationships; at this stage, **it's simply a case of determining and describing the “what”**.
- ❖ Descriptive analytics draws on descriptive statistics, which you can learn about [here](#).
- ❖ **Data aggregation is the process of summarizing a large pool of data for high level analysis. At its most basic level, it involves compiling information from a range of prescribed databases and organizing it into a simpler, easy-to-use medium, usually utilizing sum, average, mean, or median references**

❖ **Diagnostic Analytics**

- ❖ **While descriptive analytics looks at the “what”, diagnostic analytics explores the “why”.**
- ❖ When running diagnostic analytics, data analysts will first seek to identify anomalies within the data—that is, anything that cannot be explained by the data in front of them.
- ❖ **For example:**
- ❖ If the data shows that there was a sudden drop in sales for the month of March, the data analyst will need to investigate the cause.
- ❖ To do this, they’ll get on what’s known as the discovery phase, identifying any additional data sources that might tell them more about why such irregularities happened.
- ❖ Finally, the data analyst will try to uncover causal relationships—for example, looking at any events that may correlate or correspond with the decrease in sales. At this stage, data analysts may use probability theory, regression analysis, filtering, and time-series data analytics.

❖ Predictive Analytics

- ❖ **Predictive analytics tries to predict what is likely to happen in the future.**
- ❖ This is where data analysts start to come up with actionable, data-driven insights that the company can use to inform their next steps.
- ❖ Predictive analytics estimates the likelihood of a future outcome based on historical data and probability theory, and while it can never be completely accurate, it does eliminate much of the guesswork from key business decisions.
- ❖ Predictive analytics can be used to forecast all sorts of outcomes—from what products will be most popular at a certain time, to how much the company revenue is likely to increase or decrease in a given period.
- ❖ Ultimately, predictive analytics is used to increase the business's chances of “hitting the mark” and taking the most appropriate action.

❖ **Prescriptive Analytics**

- ❖ Prescriptive analytics advises on the actions and decisions that should be taken.
- ❖ In other words, prescriptive analytics shows you how you can take advantage of the outcomes that have been predicted.
- ❖ When conducting prescriptive analysis, data analysts will consider a range of possible scenarios and assess the different actions the company might take.
- ❖ Prescriptive analytics is one of the more complex types of analysis, and may involve working with algorithms, machine learning, and computational modeling procedures.
- ❖ However, the effective use of prescriptive analytics can have a huge impact on the company's decision-making process and, ultimately, on the bottom line.